

Computer Graphics Midterm Project: 3D OBJ Parser

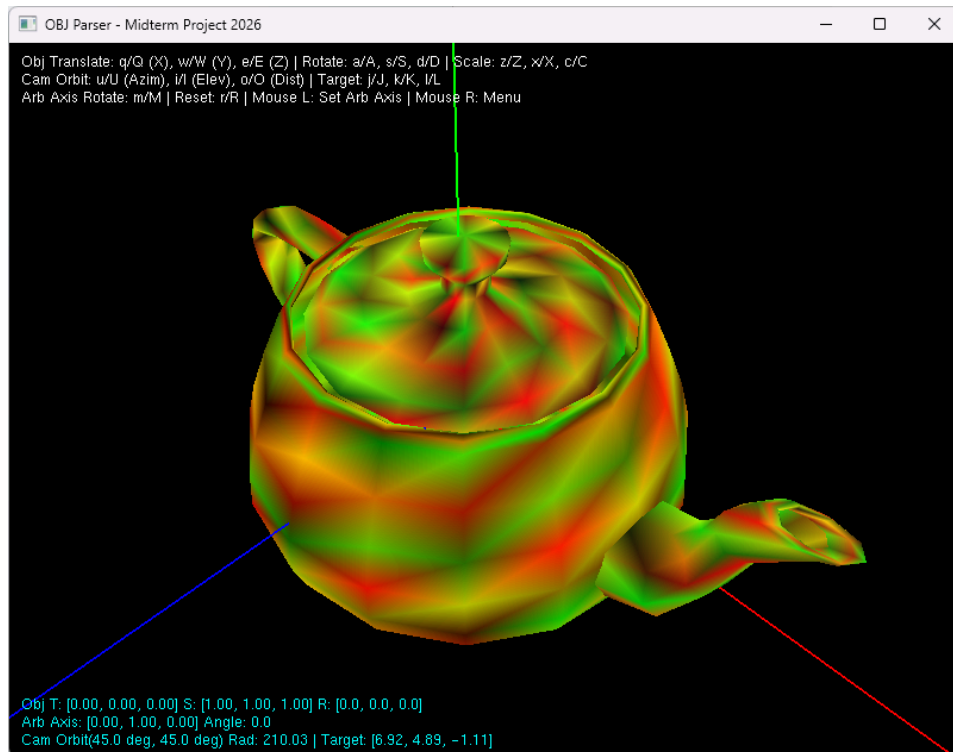
111590453 資工四 張竣崴

1. Overview

This application is a 3D model viewer designed to load and manipulate .obj files. It features a high-precision Vertex-Based Frustum-Fit system that ensures every model is automatically centered and scaled to occupy 80% of the screen height, regardless of its original coordinate system or geometry complexity. The project supports comprehensive object transformations, an orbiting camera with adjustable focus, and multiple rendering modes.

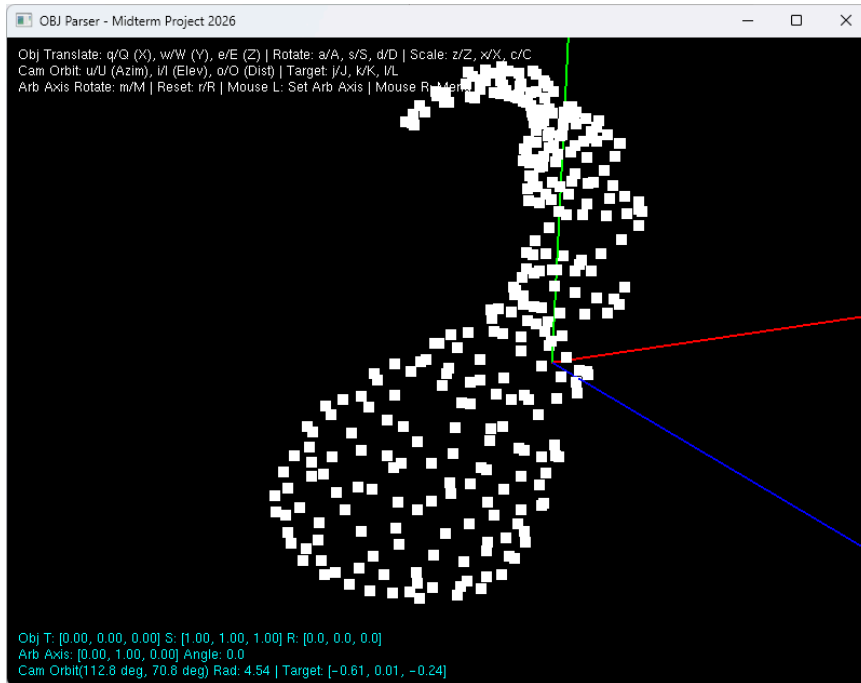
2. Loaded Objects (Screenshots)

2.1 Teapot



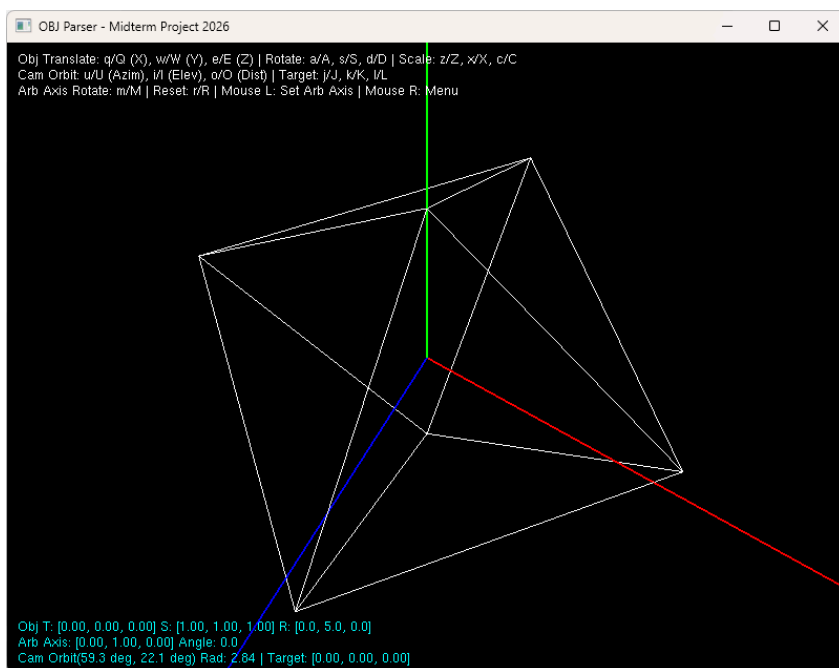
Description: The Utah Teapot rendered in face mode with the random color toggle enabled.

2.2 Gourd



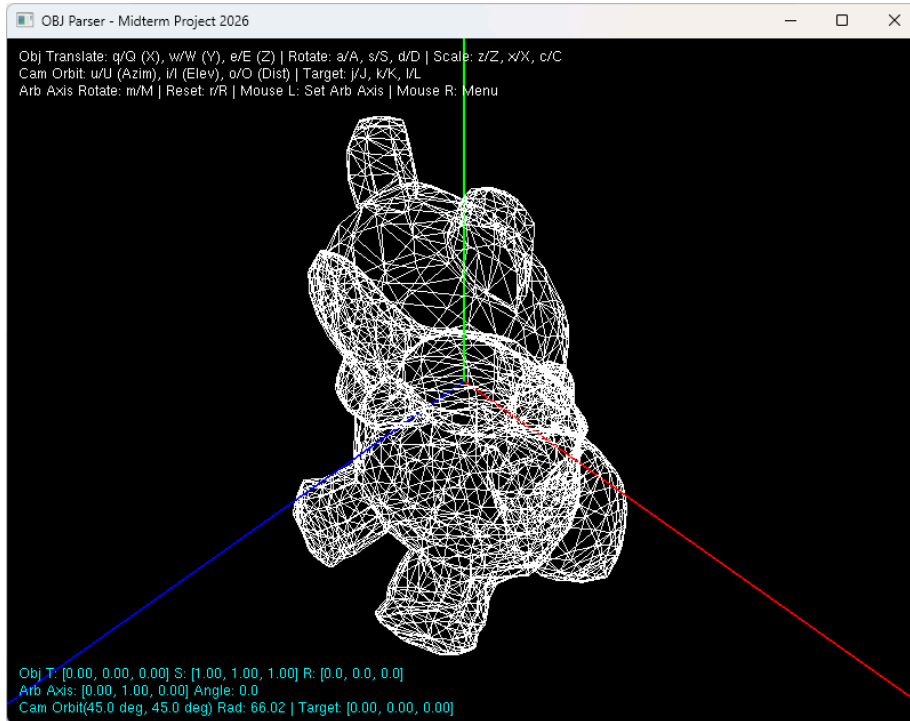
Description: The Gourd model rendered using point mode.

2.3 Octahedron



Description: A primitive octahedron rendered using line mode.

2.4 Teddy



Description: The Teddy bear rendered using line mode.

3. Control Guide

3.1 Object Transformation (SRT)

- **Translation:**
 - q / Q: Move along **X-axis**
 - w / W: Move along **Y-axis**
 - e / E: Move along **Z-axis**
- **Rotation:**
 - a / A: Rotate around **X-axis**
 - s / S: Rotate around **Y-axis**
 - d / D: Rotate around **Z-axis**
- **Scaling:**
 - z / Z: Scale **X-axis**
 - x / X: Scale **Y-axis**
 - c / C: Scale **Z-axis**

3.2 Camera Controls

- **Orbit (Rotation):**
 - u / U: Adjust **Azimuth** (Theta)
 - i / I: Adjust **Elevation** (Phi)
- **Zoom:**
 - o / O: Move camera closer or further from the target (Radius/Distance).
- **Target (Focus Point):**
 - j / J: Move target along **X-axis**
 - k / K: Move target along **Y-axis**
 - l / L: Move target along **Z-axis**

3.3 Mouse & Misc

- **Left Click:** Click to define the **Arbitrary Axis** for rotation.
- **m / M:** Rotate around the selected **Arbitrary Axis**.
- **Right Click:** Context Menu (Render Mode, File Loading, Random Color).
- **r / R: Reset View** (Re-centers model and re-runs auto-scaling).
- **Esc:** Exit.